SCHEDULE OF TECHNICAL SPECIFICATIONS (EPR/ESR PROBE)

Ref: R&D-TENDER-2026-PHYSICS-09

1. Resonator Absolute Sensitivity

Absolute sensitivity: >= 1.5 x 10^9 spins/G line width (weak pitch sample);

2. Working Frequency Range

Operational frequency band: 9.20 GHz to 9.85 GHz with 1 kHz synthesizer resolution;

3. Microwave Leveled Power

Output microwave power in leveled mode: 200 mW +/- 10 mW continuous;

4. Dynamic Attenuation Precision

Calibrated attenuation range: 0 to 60 dB in steps of 1 dB with precision +/- 0.25 dB;

5. Sample Tube Volume Capability

Sample volume compatibility: 5 uL to 100 uL capillary tubes with quartz CFQ grade;

6. Characteristic RF Impedance

Nominal microwave impedance: 50 Ohm +/- 0.5 Ohm, water-cooled circulating jacket;

7. Magnetic Field Stability

Long term field stability: +/- 10 mG/hour under room temperature variation < +/- 1.0 °C;

8. Cryogenic Temperature Bounds

Operating temperature span: 4.2 K to 350 K with temperature stability of +/- 0.05 K;

9. Modulation Linearity

Modulation frequency: 100 kHz with harmonic distortion <= -55 dBc;